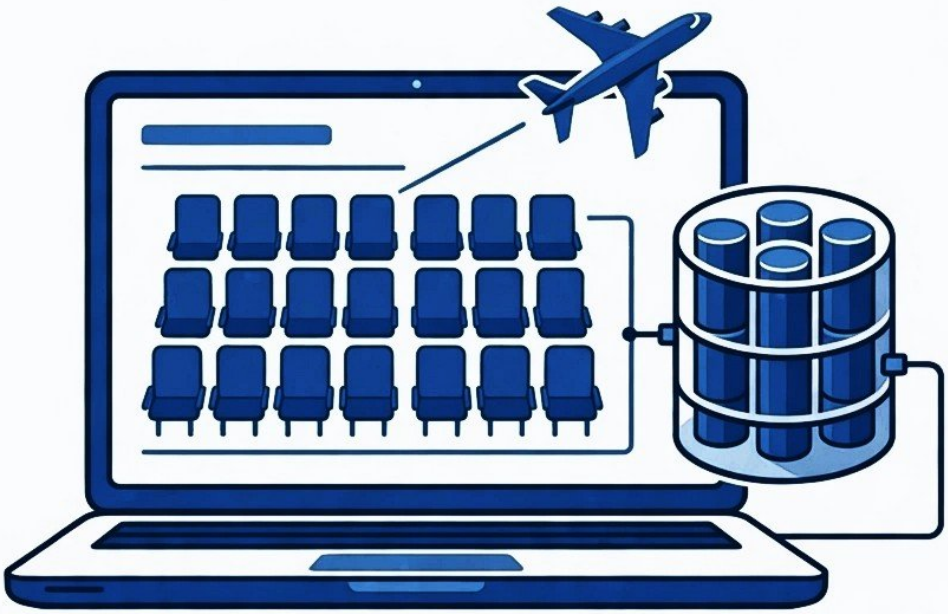
COMPUTER SCIENCE PROJECT

FLIGHT SEAT MANAGEMENT SYSTEM

Using Python and SQL Connectivity



Submitted by:

Name: Bhavneet Singh
Class & Section: XII - C
Roll Number: 6

Submitted to:

Computer Science Teacher: Ms. Neetu Mam

School Name: Kendriya Vidyalaya, Delhi Cantt

Academic Session: 2026-2027

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CERTIFICATE

This is to certify that Bhavneet Singh a student of Class XII, Section C, Roll Number 6, has successfully completed the Computer Science project entitled "Flight Seat Management System Using Python and SQL Connectivity" during the academic session 2026-27.

The project has been completed under my guidance and supervision. It demonstrates the student's understanding of Python programming, functions, conditional statements, loops and SQL connectivity using MySQL.

To the best of my knowledge, this project is the student's original work and has been completed as part of the requirements for the Class XII Computer Science Practical Examination.

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Teacher's Signature: _____

External Examiner's Signature: _____

Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my Computer Science teacher for providing me with the opportunity to complete this project entitled "Flight Seat Management System Using Python and SQL Connectivity."

I am thankful to my teacher for the valuable guidance, encouragement and support provided throughout the development of this project.

I would also like to thank my school for providing the necessary learning environment and resources.

This project helped me understand the practical application of Python programming, particularly functions, loops, conditional statements and SQL connectivity using MySQL.

Finally, I am grateful to my parents and everyone who supported and encouraged me during the completion of this project.

Name: Bhavneet Singh
Class: XII C
Roll Number: 6

Project Introduction

Flight Seat Management System

A Flight Seat Management System is a computer-based application designed to manage the records of flights and passenger bookings. It helps maintain flight information systematically and makes common flight operations like booking, cancellation and checking availability easier, faster and more reliable.

In a traditional airline office, flight records are often maintained manually in registers. This process can be time-consuming and may lead to errors, duplication of records, double booking and difficulty in searching for flights. A computerized system helps overcome these problems by storing and managing information digitally.
digitally.

This project has been developed using the Python programming language and SQL connectivity using MySQL. The MySQL database is used to store flight and booking records permanently, even after the program is closed. The system provides a menu-driven interface through which the user can add new flights, display all available flights, search for a particular flight, book a seat, cancel a booking, view passenger details and update or delete flight records. The project demonstrates the practical application of Python concepts such as functions, loops, conditional statements, database handling and the mysul.connector module.

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File Edit Format Run Options Window Help

```
import mysql.connector

mydb = mysql.connector.connect(
    host="localhost",
    port=3306,
    user="root",
    password="bhavi"
)

if mydb.is_connected():
    print("MySQL Connection Successful!")


'''mycur = mydb.cursor()

mycur.execute("CREATE DATABASE flight_management")

print("Database Created Successfully!")'''

mycur = mydb.cursor()
mycur.execute("USE flight_management")

mycur.execute("SELECT DATABASE()")
print(mycur.fetchone())

'''# CREATING THE SEATS TABLE
```

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```
'''# CREATING THE SEATS TABLE

# SQL query to create a table named seats
sql = """
CREATE TABLE seats (
    seat_id INT PRIMARY KEY,
    flight_id INT,
    seat_no VARCHAR(10),
    status VARCHAR(20)
)
"""

# Executing the SQL query
mycur.execute(sql)

# Saving the changes in the database
mydb.commit()

# Displaying success message
print("Seats Table Created Successfully!")
'''

# INSERTING DATA INTO THE SEATS TABLE

# FUNCTION FOR ADDING SEAT

def add_seat():
```

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```
def add_seat():

    # Taking input from the user
    seat_id = int(input("Enter seat ID: "))
    flight_id = int(input("Enter flight ID: "))
    seat_no = input("Enter seat number: ")
    status = input("Enter seat status: ")

    # SQL query to insert data into seats table
    sql = """
    INSERT INTO seats (seat_id, flight_id, seat_no, status)
    VALUES (%s, %s, %s, %s)
    """

    # Storing user input in a tuple
    val = (seat_id, flight_id, seat_no, status)

    # Executing the SQL query
    mycur.execute(sql, val)

    # Saving changes in the database
    mydb.commit()

    # Displaying success message
    print("Seat Added Successfully!")
```

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last.py - C:/Users/Japneet/AppData/Local/Programs/Python/Python312/last.py (3.12.10)

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```
# FUNCTION FOR MODIFYING SEAT



def modify_seat():

    # Taking seat ID from the user
    seat_id = int(input("Enter seat ID which you want to modify: "))

    # Taking new seat details
    seat_no = input("Enter new seat number: ")
    status = input("Enter new seat status: ")

    # SQL query to update seat details
    sql = """
    UPDATE seats
    SET seat_no = %s, status = %s
    WHERE seat_id = %s
    """

    # Storing values in a tuple
    val = (seat_no, status, seat_id)

    # Executing the SQL query
    mycur.execute(sql, val)

    # Saving changes in database
```

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last.py - C:/Users/Japneet/AppData/Local/Programs/Python/Python312/last.py (3.12.10)

File Edit Format Run Options Window Help

```
    # Displaying success message
    print("Seat Modified Successfully!")


# FUNCTION FOR DELETING SEAT

def delete_seat():

    # Taking seat ID from the user
    seat_id = int(input("Enter seat ID to delete: "))

    # SQL query to delete a seat
    sql = """
    DELETE FROM seats
    WHERE seat_id = %s
    """

    # Storing seat ID in a tuple
    val = (seat_id,)

    # Executing the SQL query
    mycur.execute(sql, val)

    # Saving changes in database
    mydb.commit()

    # Displaying success message
```

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last.py - C:/Users/Japneet/AppData/Local/Programs/Python/Python312/last.py (3.12.10)

File Edit Format Run Options Window Help

```
    print("Seat Deleted Successfully!")

# FUNCTION FOR DISPLAYING SEATS

def display_seat():

    # SQL query to fetch all records from seats table
    sql = """
    SELECT * FROM seats
    """

    # Executing the SQL query
    mycur.execute(sql)

    # Fetching all records from the table
    records = mycur.fetchall()

    # Displaying all records
    for i in records:
        print(i)

# MENU FOR USER

while True:

    print("       ✈     SEAT MANAGEMENT SYSTEM   ✈   ")
    print("\n----> ENTER 1 TO Add Seat")
```

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last.py - C:/Users/Japneet/AppData/Local/Programs/Python/Python312/last.py (3.12.10)

File Edit Format Run Options Window Help

```
# MENU FOR USER

while True:

    print("       ✈     SEAT MANAGEMENT SYSTEM   ✈   ")
    print("\n----> ENTER 1 TO Add Seat")
    print("----> ENTER 2 TO Modify Seat")
    print("----> ENTER 3 TO Delete Seat")
    print("----> ENTER 4 TO Display Seats")
    print("----> ENTER 5 TO Exit")

    # Taking choice from the user
    choice = int(input("Enter your choice: "))

    if choice == 1:
        add_seat()

    elif choice == 2:
        modify_seat()

    elif choice == 3:
        delete_seat()

    elif choice == 4:
        display_seat()
```

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last.py - C:/Users/Japneet/AppData/Local/Programs/Python/Python312/last.py (3.12.10)

File Edit Format Run Options Window Help

```
    print("----> ENTER 3 TO Delete Seat")
    print("----> ENTER 4 TO Display Seats")
    print("----> ENTER 5 TO Exit")

    # Taking choice from the user
    choice = int(input("Enter your choice: "))

    if choice == 1:
        add_seat()

    elif choice == 2:
        modify_seat()

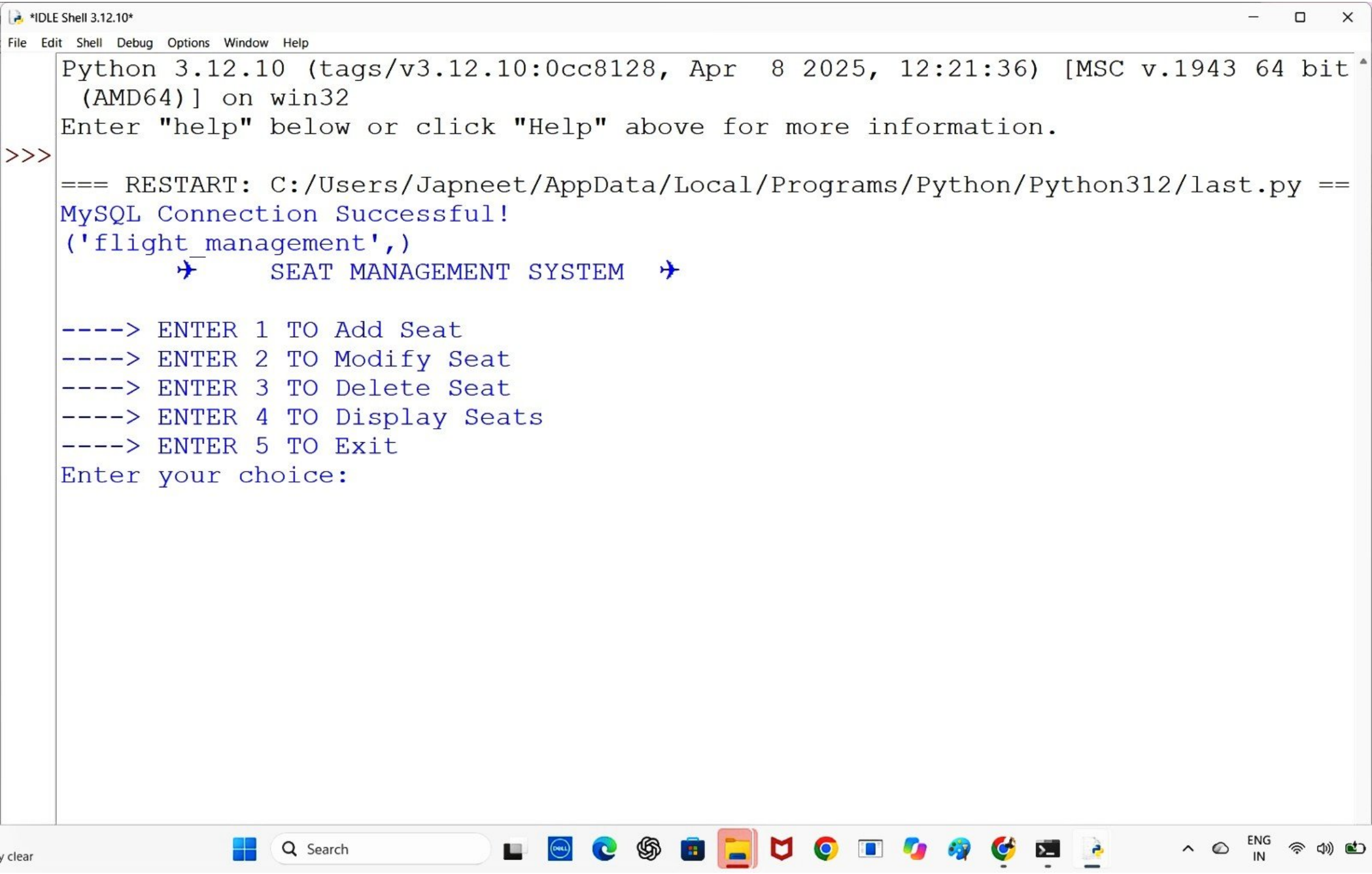
    elif choice == 3:
        delete_seat()

    elif choice == 4:
        display_seat()

    elif choice == 5:
        print("Thank You For Using Seat Management System!")
        break

    else:
        print("Invalid Choice! Please Try Again.")
```

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IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Japneet/AppData/Local/Programs/Python/Python312/last.py ==
MySQL Connection Successful!
('flight_management',)
✈ SEAT MANAGEMENT SYSTEM ✈
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice:
Search
ENG
IN

IDLE Shell 3.12.10

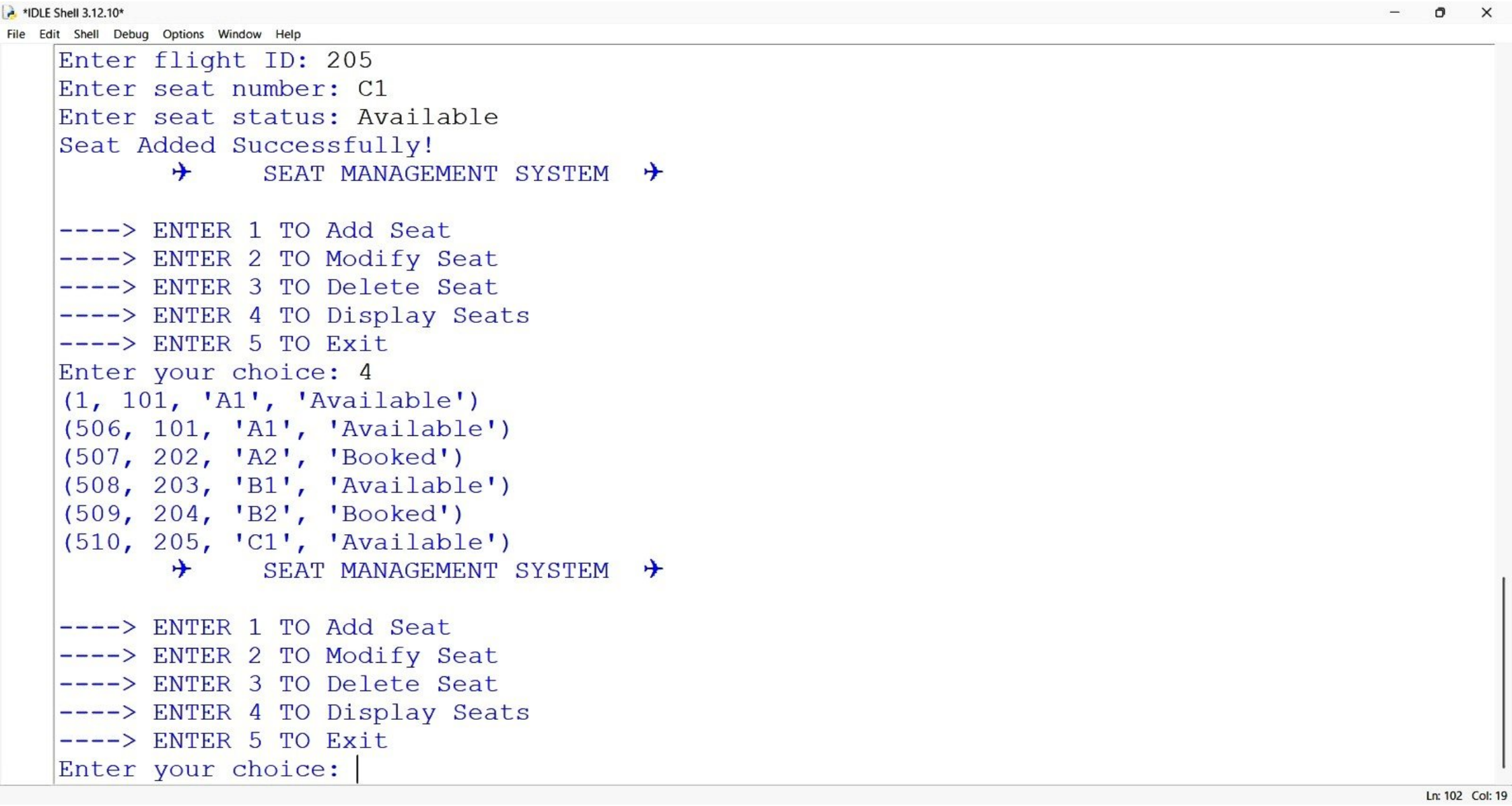
File Edit Shell Debug Options Window Help

```
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Japneet/AppData/Local/Programs/Python/Python312/last.py ==
MySQL Connection Successful!
('flight_management',)
       ✈     SEAT MANAGEMENT SYSTEM  ✈

----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 1
Enter seat ID: 506
Enter flight ID: 101
Enter seat number: A1
Enter seat status: Available
Seat Added Successfully!
       ✈     SEAT MANAGEMENT SYSTEM  ✈

----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice:
```

Ln: 27 Col: 19


IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Enter flight ID: 205
Enter seat number: C1
Enter seat status: Available
Seat Added Successfully!
✈ SEAT MANAGEMENT SYSTEM ✈
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 4
(1, 101, 'A1', 'Available')
(506, 101, 'A1', 'Available')
(507, 202, 'A2', 'Booked')
(508, 203, 'B1', 'Available')
(509, 204, 'B2', 'Booked')
(510, 205, 'C1', 'Available')
✈ SEAT MANAGEMENT SYSTEM ✈
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice:
Ln: 102 Col: 19

IDLE Shell 3.12.10

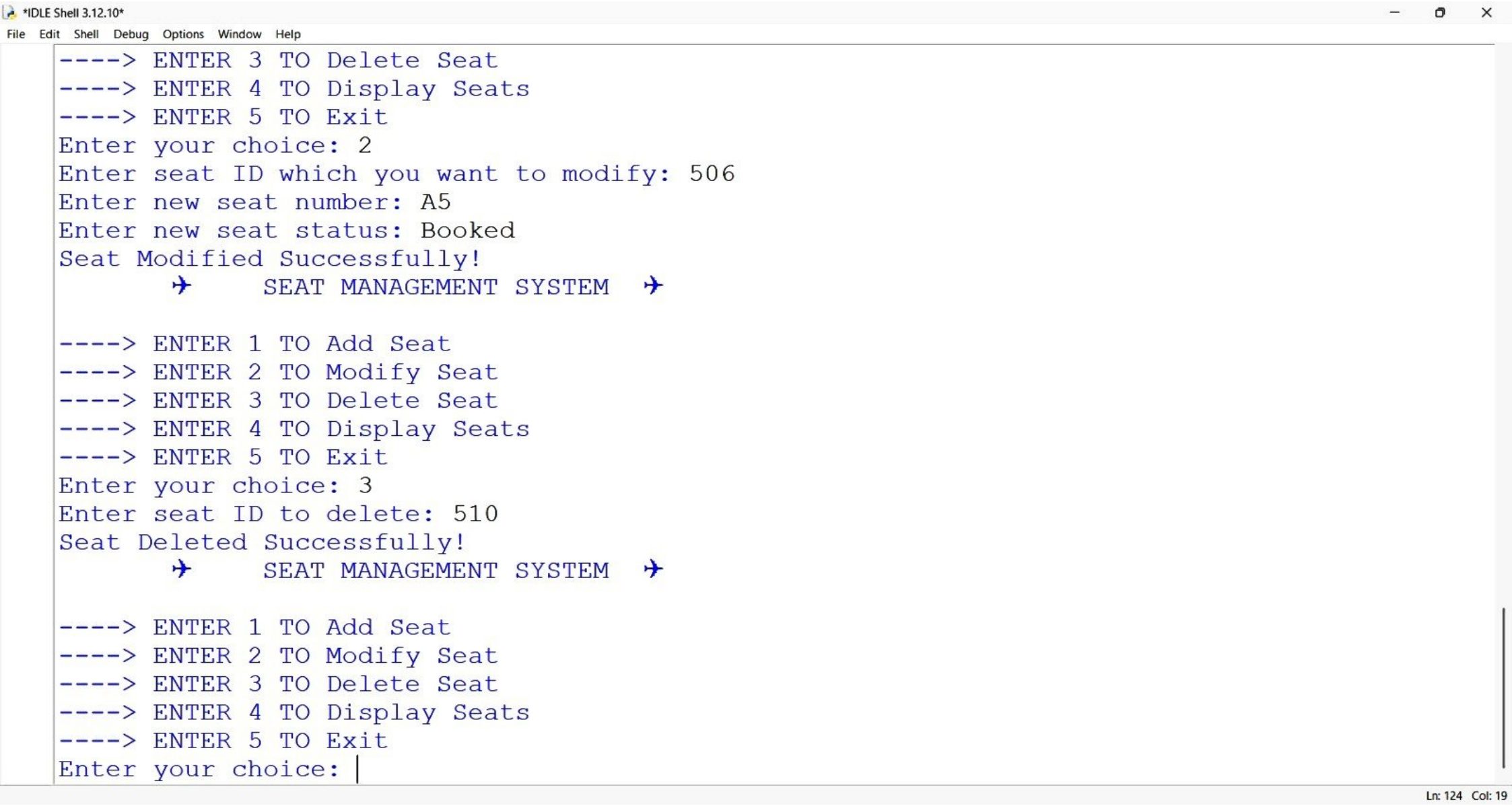
File Edit Shell Debug Options Window Help

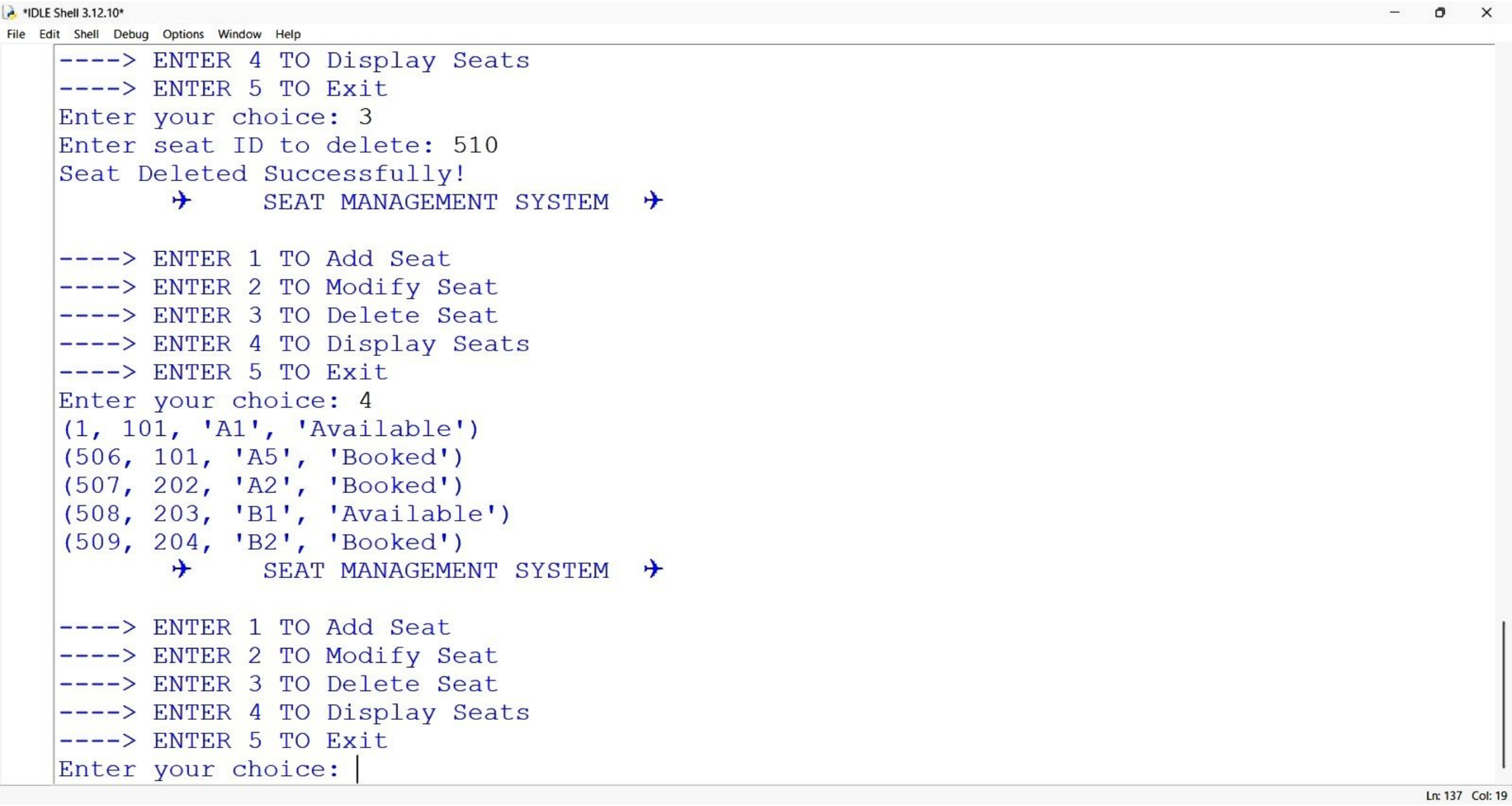
```
(1, 101, 'A1', 'Available')
(506, 101, 'A1', 'Available')
(507, 202, 'A2', 'Booked')
(508, 203, 'B1', 'Available')
(509, 204, 'B2', 'Booked')
(510, 205, 'C1', 'Available')
       ✈     SEAT MANAGEMENT SYSTEM  ✈

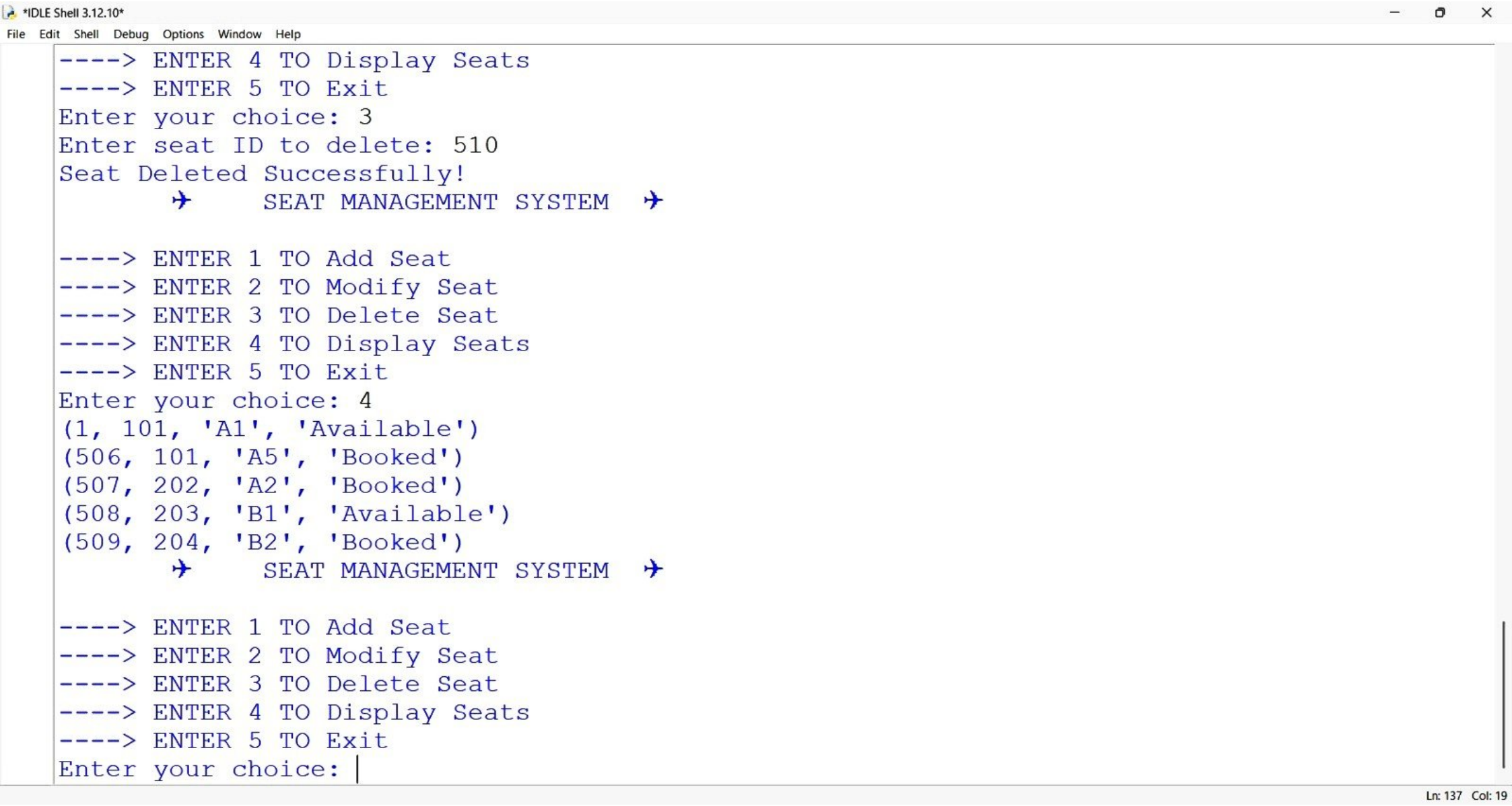
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 2
Enter seat ID which you want to modify: 506
Enter new seat number: A5
Enter new seat status: Booked
Seat Modified Successfully!
       ✈     SEAT MANAGEMENT SYSTEM  ✈

----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice:
```

Ln: 114 Col: 19


IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 2
Enter seat ID which you want to modify: 506
Enter new seat number: A5
Enter new seat status: Booked
Seat Modified Successfully!
✈ SEAT MANAGEMENT SYSTEM ✈
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 3
Enter seat ID to delete: 510
Seat Deleted Successfully!
✈ SEAT MANAGEMENT SYSTEM ✈
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice:
Ln: 124 Col: 19


IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 3
Enter seat ID to delete: 510
Seat Deleted Successfully!
✈ SEAT MANAGEMENT SYSTEM ✈
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 4
(1, 101, 'A1', 'Available')
(506, 101, 'A5', 'Booked')
(507, 202, 'A2', 'Booked')
(508, 203, 'B1', 'Available')
(509, 204, 'B2', 'Booked')
✈ SEAT MANAGEMENT SYSTEM ✈
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice:
Ln: 137 Col: 19


IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 3
Enter seat ID to delete: 510
Seat Deleted Successfully!
✈ SEAT MANAGEMENT SYSTEM ✈
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 4
(1, 101, 'A1', 'Available')
(506, 101, 'A5', 'Booked')
(507, 202, 'A2', 'Booked')
(508, 203, 'B1', 'Available')
(509, 204, 'B2', 'Booked')
✈ SEAT MANAGEMENT SYSTEM ✈
----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice:
Ln: 137 Col: 19

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Enter your choice: 3
Enter seat ID to delete: 510
Seat Deleted Successfully!
       ✈     SEAT MANAGEMENT SYSTEM  ✈

----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 4
(1, 101, 'A1', 'Available')
(506, 101, 'A5', 'Booked')
(507, 202, 'A2', 'Booked')
(508, 203, 'B1', 'Available')
(509, 204, 'B2', 'Booked')
       ✈     SEAT MANAGEMENT SYSTEM  ✈

----> ENTER 1 TO Add Seat
----> ENTER 2 TO Modify Seat
----> ENTER 3 TO Delete Seat
----> ENTER 4 TO Display Seats
----> ENTER 5 TO Exit
Enter your choice: 5
Thank You For Using Seat Management System!
>>>
```

Ln: 139 Col: 0